

TECHBIZ SECURITY

Cyber Security Internship Program

Week 3 – Task 8 (SOC)

Cyber Kill Chain + MITRE ATT&CK Introduction

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Task Objective

This report explains the Cyber Kill Chain and MITRE ATT&CK frameworks and describes how they help security professionals and SOC analysts analyze attacker behavior. The task involved studying MITRE ATT&CK technique T1110 – Brute Force, reviewing Enterprise ATT&CK tactics, and conceptually mapping a realistic attack scenario across the seven stages of the Cyber Kill Chain.

1. Introduction

The Cyber Kill Chain and MITRE ATT&CK are two widely used frameworks that help security professionals understand, classify, and analyze adversary behavior during a cyberattack. Both frameworks provide a structured way of thinking about how attacks unfold, which supports detection, investigation, and response activities within a Security Operations Center (SOC).

For a SOC analyst, these frameworks are valuable because they:

- Help analysts understand where an attacker may be in the overall attack lifecycle.
- Help classify observed attacker behavior in a consistent and standardized manner.
- Help correlate security alerts with known adversary tactics and techniques.
- Help improve detection, investigation, and incident response processes.

The Lockheed Martin Cyber Kill Chain describes the sequential stages an attacker typically progresses through, from initial reconnaissance to achieving their final objective. The MITRE ATT&CK Framework, maintained by the MITRE Corporation, is a globally accessible knowledge base of adversary tactics, techniques, and procedures (TTPs) based on real-world observations.

The key difference between the two frameworks is their focus: the Cyber Kill Chain focuses on the broad stages of an attack lifecycle, while MITRE ATT&CK focuses on the specific tactics, techniques, and procedures adversaries use during each stage of an attack.

2. Cyber Kill Chain – 7 Stages

The Cyber Kill Chain consists of seven sequential stages that describe the typical progression of a cyberattack.

Stage	Description
1. Reconnaissance	The attacker gathers information about the target, such as employees, domains, email addresses, technologies, or other publicly available information.
2. Weaponization	The attacker prepares a malicious payload, document, link, malware, or other mechanism intended to compromise the target.
3. Delivery	The attacker delivers the malicious payload or attack mechanism to the target, such as through phishing email, malicious attachment, website, or other delivery method.
4. Exploitation	The attacker takes advantage of a vulnerability, user interaction, or other condition to execute malicious activity or obtain access.

Stage	Description
5. Installation	The attacker installs malware or establishes a mechanism that allows continued access to the compromised environment.
6. Command and Control (C2)	The compromised system communicates with attacker-controlled infrastructure, allowing the attacker to control or interact with the compromised environment.
7. Actions on Objectives	The attacker performs the intended objective, such as stealing information, accessing sensitive systems, disrupting services, or abusing compromised accounts.

Memory sequence:

Reconnaissance → Weaponization → Delivery → Exploitation → Installation → Command & Control → Actions on Objectives

3. MITRE ATT&CK Framework

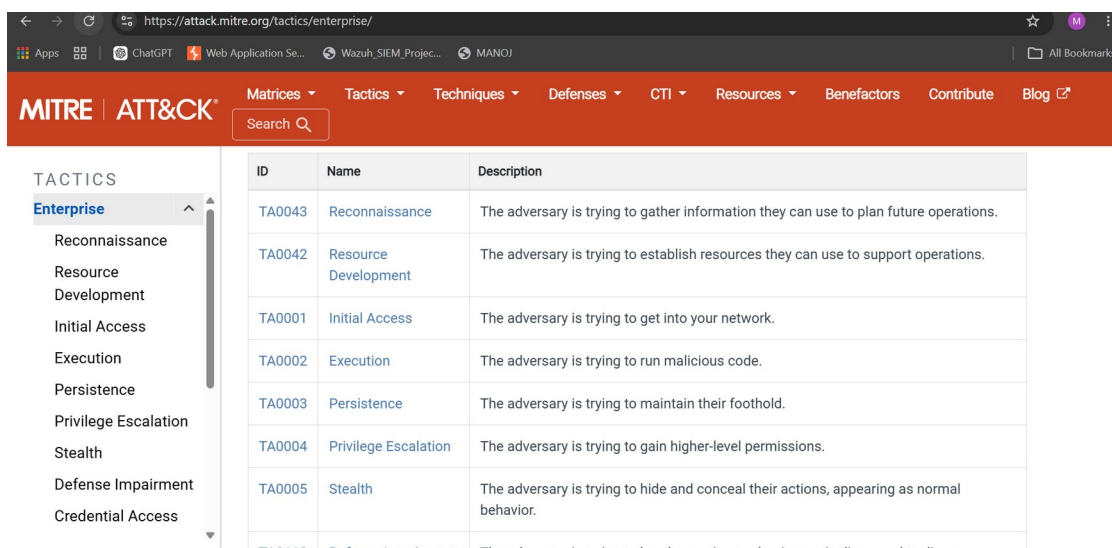
MITRE ATT&CK is a globally accessible knowledge base that organizes adversary behavior using three hierarchical levels: Tactics, Techniques, and Sub-techniques.

- Tactics describe the adversary's objective or reason for performing an activity (the "why").
- Techniques describe how the adversary attempts to achieve that objective (the "how").
- Sub-techniques provide more specific forms of a technique, describing a more granular method of achieving the same goal.

This structure allows SOC analysts and threat intelligence teams to describe adversary behavior in a consistent, standardized way that can be shared and compared across organizations.

4. MITRE ATT&CK Enterprise Tactics

The official MITRE ATT&CK website was used to review the Enterprise ATT&CK matrix, which lists the tactics adversaries use against enterprise IT environments.



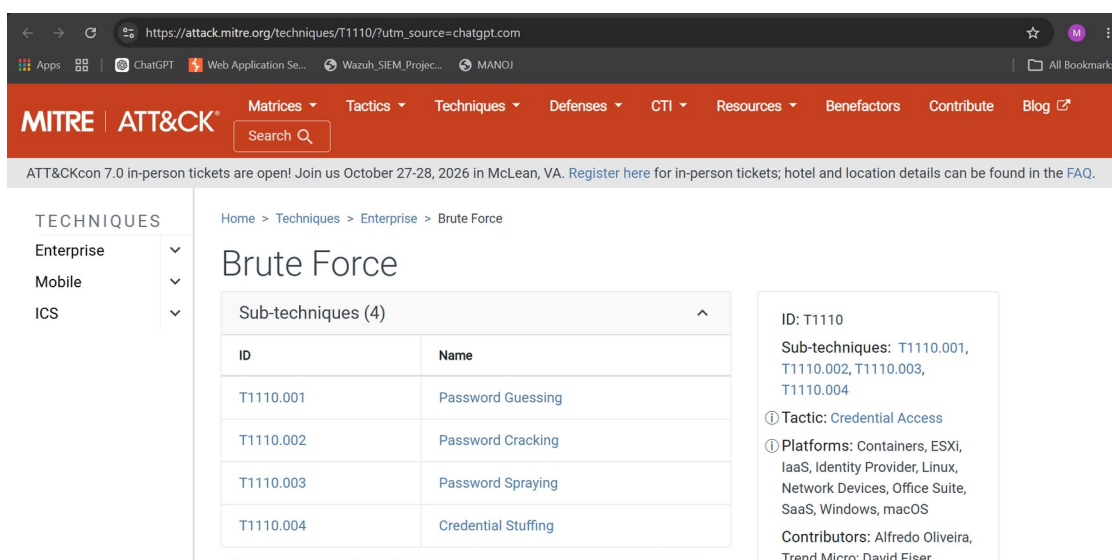
ID	Name	Description
TA0043	Reconnaissance	The adversary is trying to gather information they can use to plan future operations.
TA0042	Resource Development	The adversary is trying to establish resources they can use to support operations.
TA0001	Initial Access	The adversary is trying to get into your network.
TA0002	Execution	The adversary is trying to run malicious code.
TA0003	Persistence	The adversary is trying to maintain their foothold.
TA0004	Privilege Escalation	The adversary is trying to gain higher-level permissions.
TA0005	Stealth	The adversary is trying to hide and conceal their actions, appearing as normal behavior.
TA0112	Defense Impairment	The adversary is trying to break security mechanisms, pipelines, and tooling on

Figure 1: MITRE ATT&CK Enterprise tactics viewed on the official MITRE ATT&CK website.

The screenshot above demonstrates a portion of the Enterprise ATT&CK tactics list, including examples such as Reconnaissance, Resource Development, Initial Access, Execution, Persistence, Privilege Escalation, Stealth, Defense Impairment, and Credential Access. Each of these tactics represents a different objective that an adversary may be trying to achieve at a given stage of an attack. The screenshot reflects the visible portion of the tactics list at the time of review and does not represent every tactic in the full Enterprise matrix.

5. MITRE ATT&CK Technique – T1110 Brute Force

The official MITRE ATT&CK website was also used to study technique T1110 – Brute Force in detail.



ID	Name
T1110.001	Password Guessing
T1110.002	Password Cracking
T1110.003	Password Spraying
T1110.004	Credential Stuffing

ID: T1110
Sub-techniques: T1110.001, T1110.002, T1110.003, T1110.004
Tactic: Credential Access
Platforms: Containers, ESXi, IaaS, Identity Provider, Linux, Network Devices, Office Suite, SaaS, Windows, macOS
Contributors: Alfredo Oliveira, Trend Micro; David Fiser,

Figure 2: MITRE ATT&CK T1110 – Brute Force technique.

The screenshot shows the following evidence:

- Technique name: Brute Force
- Technique ID: T1110
- Tactic: Credential Access

- Four sub-techniques are listed under this technique.

Brute Force is a credential-access technique in which an adversary systematically attempts to obtain valid credentials when passwords are unknown or when password hashes have been obtained. This technique falls under the Credential Access tactic, meaning the adversary's objective at this stage is to steal account names and passwords.

T1110 Sub-techniques

Sub-technique	Description
T1110.001 – Password Guessing	Attempts to identify passwords by trying likely or commonly used password values.
T1110.002 – Password Cracking	Attempts to recover passwords from password hashes or other credential representations.
T1110.003 – Password Spraying	Attempts a small number of commonly used passwords against many accounts to reduce the likelihood of account lockout.
T1110.004 – Credential Stuffing	Uses previously obtained username/password combinations against other services or accounts.

Common SOC Detection Indicators

At a high level, SOC analysts may look for the following indicators when investigating potential brute-force activity:

- Repeated failed authentication attempts.
- Large numbers of authentication failures within a short time period.
- One source attempting authentication against multiple accounts.
- Multiple sources attempting authentication against the same account.
- Successful authentication occurring after many failed attempts.
- Unusual authentication patterns, such as logins at abnormal times or from unexpected locations.

6. TryHackMe MITRE Room Status

The screenshot shows the TryHackMe website interface. At the top, there's a navigation bar with links to Dashboard, Learn, Practice, and Compete. Below this, a banner reads "'MITRE' is a Premium room." with a subtext "Subscribe to join it now or explore over 500 free rooms." There are two buttons: "Individual Plans" and "Business Plans".

Below the banner, there are three pricing cards:

- Free:** Explore cyber security basics. Price: ₹0. Features include limited access to learning paths, free rooms only, and a 1hr daily limit on AttackBox.
- Premium:** Best for building core skills. Price: ₹499.00 (discounted to ₹261.97/month). It includes a "Get 6 months free with annual" offer and a "Billed annually Save 47%" indicator. A discount coupon "0252026" is applied.
- MAX:** Best for mastering advanced skills. Price: ₹4732.00 (discounted to ₹1249.92/month). It includes a "Billed annually Save 28%" indicator and an "Upgrade to MAX" button.

At the bottom right, there's a "Secure payment - Cancel anytime" note.

Figure 3: TryHackMe MITRE room showing Premium subscription requirement.

The assigned TryHackMe "MITRE" room was checked during this task; however, as shown in the screenshot above, the room is currently marked as a Premium room and requires a paid subscription to access. Since a Premium subscription was not available, the room was not completed and was not accessed using a paid account.

As an alternative learning source, the official MITRE ATT&CK website (attack.mitre.org) was used to study the required concepts, including the Enterprise ATT&CK tactics and the T1110 – Brute Force technique. This approach allowed the core learning objectives of the task to be met using a publicly available and authoritative reference source.

7. Practical Learning Performed

The following activities were carried out as part of this task:

1. Reviewed the seven stages of the Cyber Kill Chain.
2. Studied the purpose of the Cyber Kill Chain in understanding attacker progression.
3. Opened and reviewed the official MITRE ATT&CK Enterprise tactics.
4. Studied MITRE ATT&CK technique T1110 – Brute Force.
5. Reviewed the four T1110 sub-techniques.
6. Compared Cyber Kill Chain stages with MITRE ATT&CK tactics and techniques.
7. Mapped a realistic phishing-based credential theft scenario across the Cyber Kill Chain.

It should be noted that the attack mapping described in Section 8 was a conceptual security-analysis exercise performed for learning purposes only. No real phishing campaign, credential attack, or intrusive activity was conducted as part of this task.

8. Attack Mapping Using the Cyber Kill Chain

Scenario: Phishing-Based Credential Theft Attack

The following table maps a realistic phishing-based credential theft scenario across the seven stages of the Cyber Kill Chain. This mapping is conceptual and was prepared for analysis and learning purposes only.

Kill Chain Stage	Application to Scenario
1. Reconnaissance	The attacker researches the target employee or organization and collects publicly available information such as employee names, email addresses, organizational details, and technology information.
2. Weaponization	The attacker prepares a convincing phishing email and a malicious or credential-harvesting link intended to deceive the target.
3. Delivery	The phishing email is delivered to the target's inbox.
4. Exploitation	The victim interacts with the phishing content and may enter credentials into a fraudulent login page, allowing the attacker to obtain the credentials.
5. Installation	If the phishing campaign also delivers malware, the malicious software may be installed on the victim system to establish continued access. In a credential-only phishing scenario without malware, this stage may not

Kill Chain Stage	Application to Scenario
	occur.
6. Command and Control	If malware or another persistent compromise is established, the compromised system may communicate with attacker-controlled infrastructure.
7. Actions on Objectives	The attacker may use the stolen credentials or established access to reach sensitive systems, accounts, applications, or information.

It is important to note that not every real-world attack follows the Cyber Kill Chain perfectly, and not every attack necessarily contains all seven stages. This mapping is used as a learning tool to understand the possible progression of an attack rather than a strict rule that all attacks must follow.

9. Relationship Between Cyber Kill Chain and MITRE ATT&CK

Cyber Kill Chain	MITRE ATT&CK
- Describes the broad progression of an attack. - Focuses on attack lifecycle stages. - Useful for understanding where an attacker may be in the overall operation.	- Describes adversary behavior in greater detail. - Uses tactics, techniques, and sub-techniques. - Useful for detection, investigation, threat hunting, and incident response.

Both frameworks complement each other rather than compete. For example, a phishing attack can be viewed through the Cyber Kill Chain to understand its overall lifecycle, while MITRE ATT&CK can be used to categorize the specific adversary behaviors and techniques associated with that attack.

10. SOC Analyst Relevance

Knowledge of the Cyber Kill Chain and MITRE ATT&CK supports several core SOC analyst responsibilities, including:

- Alert investigation
- Attack-stage identification
- Threat detection
- Incident investigation
- Threat hunting
- MITRE ATT&CK mapping
- Detection engineering
- Incident response
- Understanding attacker behavior

Practical example: If a SOC analyst observes repeated authentication failures followed by a successful login, the analyst may investigate whether the behavior is consistent with brute-force activity and consider MITRE ATT&CK T1110 – Brute

Force as a relevant technique during the investigation. No actual SOC alert was generated during this task; this example is used only to illustrate how the framework would be applied in a real SOC environment.

11. Key Learnings

- Understood the seven stages of the Cyber Kill Chain.
- Learned how the Cyber Kill Chain represents attacker progression.
- Understood the purpose of MITRE ATT&CK tactics and techniques.
- Studied T1110 – Brute Force.
- Learned the four T1110 sub-techniques.
- Understood the relationship between Credential Access and Brute Force.
- Reviewed Enterprise ATT&CK tactics.
- Learned how attacks can be mapped to the Cyber Kill Chain.
- Understood how these frameworks support SOC investigation and threat detection.
- Learned the importance of documenting attack behavior using standardized frameworks.

12. Conclusion

This task provided a practical understanding of two important cybersecurity frameworks used in security operations. The Cyber Kill Chain helps visualize the progression of an attack from reconnaissance to actions on objectives, while MITRE ATT&CK provides a detailed knowledge base of adversary tactics, techniques, and sub-techniques. The T1110 – Brute Force technique studied during this task is categorized under the Credential Access tactic and represents a common method attackers use to obtain valid credentials.

Framework-based analysis of this kind helps SOC analysts investigate alerts more effectively and develop a stronger understanding of attacker behavior. The phishing-based credential theft mapping exercise demonstrated how a realistic attack scenario can be analyzed across the seven stages of the Cyber Kill Chain, reinforcing the practical value of both frameworks in a SOC environment.

13. Evidence Summary

Filename	Description
01_mitre_t1110_brute_force.png	Official MITRE ATT&CK T1110 Brute Force technique page showing the technique ID, Credential Access tactic, and sub-techniques.
02_mitre_attack_enterprise_tactics.png	Official MITRE ATT&CK Enterprise tactics page showing Enterprise tactics and their descriptions.
03_tryhackme_mitre_room_premium_notice.png	TryHackMe MITRE room page showing that the room requires Premium subscription.